

## ALEXANDROUPOLI, Greece

WMO: 166270

Lat: 40.856N

Lon: 25.956E

Elev: 24

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-17

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 23.6       | 26.8 | 14.3                          | 11.4 | 27.0 | 18.0 | 13.6 | 31.1 | 34.0                  | 34.0 | 30.8 | 43.1 | 8.0                   | 20   | 0.471 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.0                   | 94.7            | 70.9 | 91.6 | 70.3 | 89.5 | 69.5 | 74.4                | 87.2 | 72.8 | 86.0 | 71.6 | 84.7 | 12.4                 | 60   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.1                            | 110.8 | 81.9 | 68.3 | 104.0 | 80.4 | 66.6 | 98.0 | 78.6 | 38.0          | 87.5 | 36.5 | 85.8 | 35.3 | 84.5 | 81.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 24.5              | 21.1 | 18.7 | 18.0                       | 98.1 | 3.7                | 3.3 | 15.3                                               | 100.4 | 13.1       | 102.4 | 11.0       | 104.2 | 8.3        | 106.6 |
|                   |      |      | 16.4                       | 77.3 | 3.6                | 1.7 | 13.8                                               | 78.6  | 11.7       | 79.5  | 9.6        | 80.5  | 7.0        | 81.7  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 58.9   | 41.4  | 42.9  | 47.5  | 54.9  | 64.5  | 72.7  | 77.9  | 78.1  | 69.6  | 60.4  | 51.5  | 44.2  |
|                                                                        | DBStd       | 14.39  | 7.23  | 7.03  | 6.22  | 5.49  | 4.96  | 5.21  | 4.01  | 4.32  | 5.06  | 5.98  | 7.02  | 8.37  |
|                                                                        | HDD50       | 923    | 281   | 212   | 124   | 19    | 0     | 0     | 0     | 0     | 0     | 4     | 63    | 220   |
|                                                                        | HDD65       | 3503   | 733   | 618   | 544   | 304   | 70    | 4     | 0     | 0     | 17    | 163   | 405   | 645   |
|                                                                        | CDD50       | 4167   | 13    | 14    | 45    | 166   | 449   | 680   | 865   | 870   | 589   | 327   | 109   | 40    |
|                                                                        | CDD65       | 1272   | 0     | 0     | 0     | 1     | 54    | 234   | 400   | 405   | 156   | 21    | 1     | 0     |
|                                                                        | CDH74       | 12877  | 0     | 0     | 0     | 16    | 437   | 2292  | 4453  | 4399  | 1189  | 91    | 0     | 0     |
|                                                                        | CDH80       | 5061   | 0     | 0     | 0     | 2     | 65    | 790   | 1943  | 1976  | 278   | 7     | 0     | 0     |
|                                                                        | WSAvg       | 7.3    | 7.9   | 7.8   | 8.0   | 6.8   | 6.4   | 6.3   | 7.5   | 7.5   | 6.9   | 7.5   | 6.8   | 8.2   |
|                                                                        | PrecAvg     | 21.8   | 2.5   | 2.4   | 2.0   | 1.5   | 1.4   | 1.1   | 0.7   | 0.4   | 1.2   | 1.9   | 2.9   | 3.5   |
| (15) Precipitation                                                     | PrecMax     | 31.3   | 7.2   | 7.3   | 6.2   | 3.3   | 4.3   | 4.3   | 2.6   | 2.7   | 6.0   | 6.2   | 9.1   | 10.0  |
|                                                                        | PrecMin     | 13.2   | 0.2   | 0.1   | 0.1   | 0.2   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 0.0   |
|                                                                        | PrecStd     | 4.5    | 1.8   | 1.7   | 1.5   | 0.9   | 1.0   | 1.1   | 0.6   | 0.6   | 1.5   | 1.7   | 1.9   | 2.3   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 61.4  | 61.1  | 66.6  | 76.2  | 85.6  | 95.3  | 98.2  | 99.0  | 89.9  | 80.7  | 71.6  | 66.5  |
|                                                                        |             | MCWB   | 56.9  | 55.3  | 55.7  | 61.4  | 64.3  | 71.5  | 71.5  | 71.8  | 67.7  | 66.4  | 63.5  | 61.8  |
|                                                                        | 2%          | DB     | 58.8  | 59.0  | 62.9  | 71.5  | 81.0  | 91.1  | 93.7  | 95.0  | 86.1  | 76.8  | 67.9  | 62.4  |
|                                                                        |             | MCWB   | 54.9  | 54.0  | 55.3  | 59.1  | 64.6  | 70.5  | 70.2  | 71.3  | 67.5  | 64.8  | 62.1  | 58.5  |
|                                                                        | 5%          | DB     | 56.1  | 57.1  | 60.8  | 68.1  | 78.8  | 87.8  | 91.5  | 91.8  | 82.8  | 73.6  | 64.8  | 59.4  |
|                                                                        |             | MCWB   | 52.2  | 52.6  | 54.4  | 57.5  | 64.2  | 68.9  | 70.2  | 70.4  | 66.9  | 63.7  | 59.6  | 56.5  |
|                                                                        | 10%         | DB     | 53.4  | 55.0  | 58.7  | 65.8  | 75.6  | 84.4  | 89.4  | 89.6  | 80.7  | 71.2  | 62.7  | 57.2  |
|                                                                        |             | MCWB   | 49.9  | 50.7  | 53.0  | 56.5  | 63.4  | 67.9  | 69.4  | 69.6  | 66.4  | 62.6  | 58.0  | 54.2  |
|                                                                        | 0.4%        | WB     | 57.6  | 56.4  | 58.6  | 62.9  | 69.7  | 75.3  | 76.2  | 76.5  | 73.6  | 70.0  | 64.8  | 62.2  |
|                                                                        |             | MCDB   | 60.8  | 60.0  | 63.9  | 72.3  | 78.5  | 87.0  | 90.1  | 88.8  | 81.3  | 75.5  | 69.1  | 66.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 55.2  | 54.7  | 56.6  | 60.7  | 67.4  | 72.7  | 74.1  | 74.6  | 71.4  | 67.4  | 62.7  | 59.0  |
|                                                                        |             | MCDB   | 58.2  | 58.1  | 61.3  | 68.1  | 76.5  | 85.5  | 87.5  | 87.7  | 79.5  | 73.0  | 66.9  | 61.8  |
|                                                                        | 5%          | WB     | 52.8  | 52.9  | 55.1  | 59.2  | 65.7  | 71.0  | 72.5  | 72.9  | 69.6  | 65.6  | 60.6  | 57.0  |
|                                                                        |             | MCDB   | 55.8  | 56.5  | 59.5  | 66.2  | 75.1  | 83.8  | 86.4  | 86.8  | 78.7  | 71.2  | 64.3  | 59.7  |
|                                                                        | 10%         | WB     | 50.2  | 51.0  | 53.5  | 57.7  | 64.2  | 69.5  | 71.1  | 71.5  | 67.8  | 63.4  | 58.8  | 54.4  |
|                                                                        |             | MCDB   | 52.7  | 54.2  | 58.0  | 64.2  | 74.0  | 81.9  | 85.1  | 85.7  | 78.2  | 69.6  | 62.3  | 57.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.7  | 14.2  | 15.5  | 17.6  | 19.6  | 20.2  | 21.0  | 21.3  | 19.5  | 16.3  | 14.9  | 12.2  |
|                                                                        |             | MCDBR  | 14.0  | 15.9  | 17.2  | 21.3  | 23.6  | 23.5  | 23.9  | 24.5  | 22.5  | 19.1  | 16.0  | 11.3  |
|                                                                        |             | MCWBR  | 11.5  | 12.3  | 11.9  | 12.6  | 12.2  | 10.7  | 10.5  | 10.2  | 10.8  | 11.2  | 11.8  | 9.8   |
|                                                                        | 5% WB       | MCDBR  | 12.9  | 14.6  | 15.7  | 18.5  | 20.7  | 21.2  | 21.6  | 22.1  | 19.8  | 16.5  | 14.7  | 10.5  |
|                                                                        |             | MCWBR  | 11.2  | 12.0  | 11.9  | 12.3  | 12.1  | 11.4  | 11.4  | 11.2  | 11.8  | 11.8  | 11.7  | 9.5   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.364 | 0.396 | 0.440 | 0.485 | 0.458 | 0.452 | 0.450 | 0.456 | 0.443 | 0.417 | 0.389 | 0.373 |
|                                                                        | taud        |        | 2.330 | 2.219 | 2.107 | 2.003 | 2.159 | 2.220 | 2.230 | 2.204 | 2.212 | 2.255 | 2.279 | 2.289 |
|                                                                        | Ebn at Noon |        | 245   | 253   | 254   | 251   | 262   | 263   | 262   | 257   | 251   | 242   | 232   | 229   |
|                                                                        | Edh at Noon |        | 28    | 36    | 45    | 54    | 47    | 44    | 44    | 44    | 41    | 35    | 30    | 28    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 534   | 798   | 1192  | 1622  | 2054  | 2260  | 2351  | 2089  | 1553  | 1008  | 645   | 473   |
|                                                                        | RadStd      |        | 68    | 76    | 106   | 169   | 125   | 154   | 100   | 74    | 100   | 94    | 65    | 69    |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | +1.37 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -328  | +371  | +225  |
| (47) | Regional (14 neighbors) | +0.92 | N/A     | N/A    | N/A     | +0.88 | +0.83 | -115        | -259  | +209  | +106  |

Nomenclature: See separate page